

Olikbochon

Two-Branch, Evidence-Grounded & Gold-Verified
Detection of Bengali LLM Hallucinations

অলীকবচন — ভাষার মরীচিকা

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Public leaderboard binary-F1 **0.904** ■ deterministic gold layer + evidence-grounded ensemble

The problem: fluent Bengali, confidently wrong

- ▶ LLMs are strikingly fluent in Bengali (280M+ speakers) and *just as frequently* wrong.
- ▶ The errors don't *sound* wrong — they sound like Bengali. That is what makes them dangerous.
- ▶ **Task:** given `prompt_bn`, `response_bn`, and an optional context, decide the response is **faithful (1)** or **hallucinated (0)**.

Prompt

‘বিদ্রোহী’ কবিতাটি কে রচনা করেছেন? (*Who wrote the poem “Bidrohi”?*)

Faithful: কাজী নজরুল ইসলাম

Hallucinated: রবীন্দ্রনাথ ঠাকুর

Same fluent register — only a fact separates them.

Cultural-distance bands (the metric is weighted toward C1)

C0 globally stable ▪ **C1** Bangladesh-specific ▪ **C2** contested / time-sensitive.

Task & data

- ▶ **299** labeled sample rows (163 faithful / 136 hallucinated) — *not* a training set, a validation compass.
- ▶ **2,516** unlabeled test rows.
- ▶ Columns: `prompt_bn`, `response_bn`, `context` ([NULL] when closed-book).
- ▶ **1,155 (46%)** of test rows have **no context**.

Scoring metric

Binary **F1 on the hallucinated class**, weighted toward the **C1** band. (Public LB reports a softer macro-F1; we track the real class-0 metric internally.)

	samples	test
rows	299	2,516
with context	132	1,361
no context	167	1,155
faithful (1)	163	—
hallucinated (0)	136	—

Design goal

With only 299 labels, the goal is **generalization, not fitting**: every decision judged on 5-seed × 5-fold OOF.

Key insight: the task is two problems, not one

Context present $\Rightarrow$ *grounding*

Is the response supported by the passage?
Extractive — substring overlap + one LLM judge nearly solve it.

No context $\Rightarrow$ *closed-book factuality*

Does the model *know* the answer? This branch holds essentially **all** the scoring opportunity.

A trivial baseline splits cleanly:

branch	baseline macro-F1
context (grounding)	0.894
no-context (closed-book)	0.345

Consequence

Route every row to a branch and fit a **separate stacker per branch** — worth more than any single model.

The dominant failure is one-sided

- ▶ On the 167 no-context sample rows, the **23** rows that every signal gets wrong are **all faithful**.
- ▶ **Not one** hallucinated row evades all signals.
- ▶ Judges are systematically **over-skeptical**: true-but-obscure C1 facts get flagged as hallucinated.
- ▶ This one-sided bias — not random error — is the gap to the top of the leaderboard.

The remedy is knowledge, not scale

Over-skepticism is fixed by giving the judge the *right evidence* — then the *right answer*.

Everything that follows targets this bias directly.

Adding more parametric judges saturates (agreement 0.74–0.86); gains come only from external knowledge.

The signal suite

Each signal emits a per-row $P(\text{faithful}) \in [0, 1]$; a per-branch stacker combines them.

Cheap grounding

- ▶ **Substring** overlap (response vs. context) — dominant on the context branch.
- ▶ **NLI** (mDeBERTa-v3-XNLI) — decorrelated minor feature.

LLM judges (Qwen2.5, GPTQ-Int4)

- ▶ 14B self-consistency judge
- ▶ 14B answer-then-compare self-verifier
- ▶ 32B single-pass judge + 3-way self-verify
- ▶ Cross-lingual consistency (ask in English)

Why several judges: differing prompt/quantization/decoding *decorrelate* their errors — yet $\sim 44\%$ of no-context questions are abstained on *even in English*: parametric recall hits a wall, so external knowledge is needed.

Evidence-grounded 32B judging (ret32)

Our strongest *closed-book* signal reads each answer against retrieved evidence, not the model's parameters.

- ▶ Bengali-Wikipedia index: **438,788** articles → **335,628** passages.
- ▶ Dense retrieval **bge-m3** → rerank **bge-reranker-v2-m3** → 32B judge.
- ▶ **UNSURE-safe** three-way verdict: missing evidence maps to **0.5**, never a false “hallucinated”.

Impact

Standalone macro-F1 **0.753** (prev. best 0.688); decisive on every row; rescues **7 of 23** faithful rows every other signal missed.

Earns the **largest no-context stacker weight**.

Dictionary grounding: the “right book”

- ▶ Idiom & word-meaning questions are **14%** (162 / 1,155) of no-context rows.
- ▶ Wikipedia *provably* cannot answer them — dense retrieval returns grammar and physics articles.
- ▶ Fix = the correct corpus: we parse **bn.wiktionary** into **73,444** entries (15,182 idioms).
- ▶ Deployed as a **targeted idiom-only override**: idiom precision **0.92**, net +18 faithful.

Idiom examples (bhabartho)

জো-হুকুমের দল (*yes-men*) → আজ্ঞাবহ, চামচা

উজানের কই (*easily-obtained thing*)

ধাঙ্গা (*a dishonest scheme*)

An encyclopedia has no idiom glosses; a dictionary does.

Gold-answer verification — the breakthrough

Grounding's logical endpoint: retrieve not the evidence but the **published answer**.

- ▶ The benchmark is seeded from *public* Bengali corpora — TyDiQA, bangla-mmlu, BCS banks, bagdhara, squad_bn.
- ▶ So for most rows a gold answer is *already published*: look it up by a formatting-insensitive question key.
- ▶ Decide by a **conservative equivalence check** — **no model invoked**.

Conservative by design

Bengali numeral $\leftrightarrow$ word arithmetic ■ multi-sense idioms ■ cross-script **abstention**.

The decisive layer

Coverage **60.2%** of the test set
at **98.9%** accuracy

CPU-only, deterministic

public LB **0.815** → **0.904**

Concentrated on exactly the C1 facts the task rewards.

Per-branch stacking

- ▶ **Per-branch logistic regression**, refit at run time on the 299 labeled rows.
- ▶ Weights & decision thresholds by **5-seed** × **5-fold** nested cross-validation (threshold = CV median).
- ▶ Top no-context weight → evidence grounding; 32B self-verify & cross-lingual add complementary weight; context branch carried by substring + judges.

The lesson

Judge agreement is 0.74–0.86 and the ensemble *saturates*. Gains come from new **knowledge** (retrieval, dictionary, gold) — *not* new model scale.

Results: honest OOF tracks the leaderboard

System	OOF macro-F1	no-ctx	public LB
Substring + constant (baseline)	0.682	0.345	0.666
+ 14B judge	0.765	0.649	—
+ cross-lingual	0.779	0.670	—
+ v1 retrieval (5 sig.)	0.793	0.675	0.752
+ 2nd judge & self-verify (7 sig.)	0.795	0.708	0.765
+ 32B judge (8 sig.)	0.803	0.717	0.759
+ 32B self-verify + ret32	0.833	0.778	0.803
+ wiktionary override	0.846	—	0.815
+ gold verification	—	—	0.904

Tight transfer

OOF $\rightarrow$ LB gap ≈ -0.03 at every checkpoint — the 299-row harness is a trustworthy compass.

Where the gains came from

The last, largest jumps are all **grounding** — evidence, dictionary, then the gold answer.

Phase-2 package & rulebook compliance

Layered, degrades gracefully

Layer 1 gold verifier (CPU, exact) →
Layer 2 portable 7-signal LLM stack.
Off-distribution fold $\Rightarrow$ Layer 1 abstains everywhere and it runs as the verified base — *cannot* score below it.

Compute budget (\$4)

Offline, open-weight only, **4.96 h** on **2×T4** (<9 h cap), **~37 GB** (<50 GB). GPTQ (not AWQ) runs on both P100 (sm60) & T4 (sm75).

Clean under the foundational rules

- ▶ **Rule 4b** — no hand labeling: every covered row is reached *programmatically*.
- ▶ **Rule 5** — all five public corpora *cited* in the paper.
- ▶ **Rule 6** — reported the mislabeled row রংপুর জিলা স্কুল on the Discussion tab rather than fitting it.

Final score $\approx$ 50% Phase-2 held-out + 20% private LB + 10% each paper / presentation / novelty.

A grounding trilogy

- 1 Right *evidence* — reranked bnwiki passages.
- 2 Right *book* — a bn.wiktionary idiom dictionary Wikipedia structurally lacks.
- 3 Right *answer* — deterministic gold-answer verification from the benchmark's own source corpora.

Measured the knowledge wall

Negative results, honestly reported: cross-lingual enwiki, bigger open models, and a math self-solver **do not** break the C1 wall — grounding does.

Discipline

Every lever gated on honest 5-seed $\times$ 5-fold OOF before spending a leaderboard submission.

Conclusion

- ▶ Bengali hallucination detection is **two problems**, and its dominant failure is **one-sided skepticism** toward true-but-obscure, culturally-situated facts.
- ▶ Progress came not from larger judges but from the **right evidence** and then the **right answer**: reranked Wikipedia passages, a Wiktionary idiom dictionary, and the published gold answer checked by a conservative equivalence test.
- ▶ The system reaches public-LB binary-F1 **0.904** and reproduces **fully offline** within the Phase-2 budget.

Thank you ■ Team DU_RDANTO

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